**Math 241, F18** **Name:** Answer

**Quiz 11**

<u>Instructions:</u> **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.**

**Problem 1.** (5 points) Evaluate the iterated integral $\int_0^1 \int_0^z \int_0^y 2ze^{-y^2}\, dxdydz$.

$$\int_0^1\int_0^z\int_0^y 2ze^{-y^2}dx\,dy\,dz$$

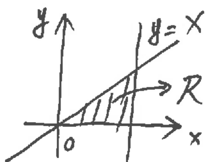
$$= \int_0^1\int_0^z 2ze^{-y^2}[x]_{x=0}^{x=y}\,dy\,dz = \int_0^1\int_0^z 2zy\,e^{-y^2}dy\,dz$$

$$= \int_0^1 z\left[-e^{-y^2}\right]_{y=0}^{y=z}dz = \int_0^1 z\left[-e^{-z^2}+1\right]dz$$

$$= \int_0^1 -ze^{-z^2}+z\,dz = \left[\frac{e^{-z^2}}{2}+\frac{z^2}{2}\right]_{z=0}^{z=1}$$

$$= \left[\frac{e^{-1}}{2}+\frac{1}{2}\right]-\left[\frac{1}{2}+0\right] = \frac{e^{-1}}{2}$$

**Problem 2.** (5 points) Evaluate $\iiint\limits_D 6xy\, dV$ where $D$ lies under the plane $z = 1 + x + y$ and above the region in the $xy$-plane enclosed by the curves $y = x$, $y = 0$, and $x = 1$.




$$R = \{(x,y) \mid 0 \le x \le 1,\ 0 \le y \le x\}$$

$$D = \{(x,y,z) \mid (x,y)\in R,\ 0 \le z \le 1+x+y\}$$

$$\Rightarrow \iiint\limits_D 6xy\,dV = \iint\limits_D \left[\int_0^{1+x+y} 6xy\,dz\right]dA$$

$$= \int_0^1\int_0^x\int_0^{1+x+y} 6xy\,dz\,dy\,dx = \int_0^1\int_0^x 6xy[z]_{z=0}^{z=1+x+y}dy\,dx$$

$$= \int_0^1\int_0^x 6xy(1+x+y)\,dy\,dx = \int_0^1\int_0^x 6xy+6x^2y+6xy^2\,dy\,dx$$

$$= \int_0^1 \left[3xy^2+3x^2y^2+2xy^3\right]_{y=0}^{y=x}dx = \int_0^1 3x^3+5x^4\,dx$$

$$= \left[\frac{3}{4}x^4+x^5\right]_{x=0}^{x=1} = \frac{3}{4}+1 = \frac{7}{4}$$